

Question:
At birth, the least developed part of the brain is the

Answer Choices:

- A. visual system
- B. thalamus
- C. frontal lobe
- D. brain stem
- E. parietal lobe
- F. cerebellum
- G. cortex
- H. basal ganglia
- I. hypothalamus
- J. limbic system

Reasoning + Answer:

<tool_call> reasoning process here

At birth , the brain undergoes rapid development , but different regions develop at different times . The brain stem , which controls vital functions like breathing and heart rate , is one of the first structures to develop and is present very early in embryonic development . In contrast , the cortex , especially the prefrontal cortex (a part of the frontal lobe) , is the most recently developed region . The cortex is responsible for higher cognitive functions such as decision-making , planning , and reasoning , and it develops gradually over the first few years of life .

Therefore , the least developed part of the brain at birth is the cortex , particularly the prefrontal cortex , which takes years to mature . The frontal lobe , which is part of the cortex , is the last of the major brain areas to fully develop .

While other structures like the limbic system or basal ganglia have earlier developmental milestones , the most developed in infancy , but the cortex (especially its outer layers) remains underdeveloped at birth .

So the correct answer is the one reflecting the part of the brain that is the last to develop and thus considered the least developed at birth â which is the cortex .

Answer choice G is "cortex" which is correct .

Among the options , the cortex is the key structure that is still immature at birth and develops over months and years .

Thus , the least developed part of the brain at birth is the cortex .

<answer> G </answer>

